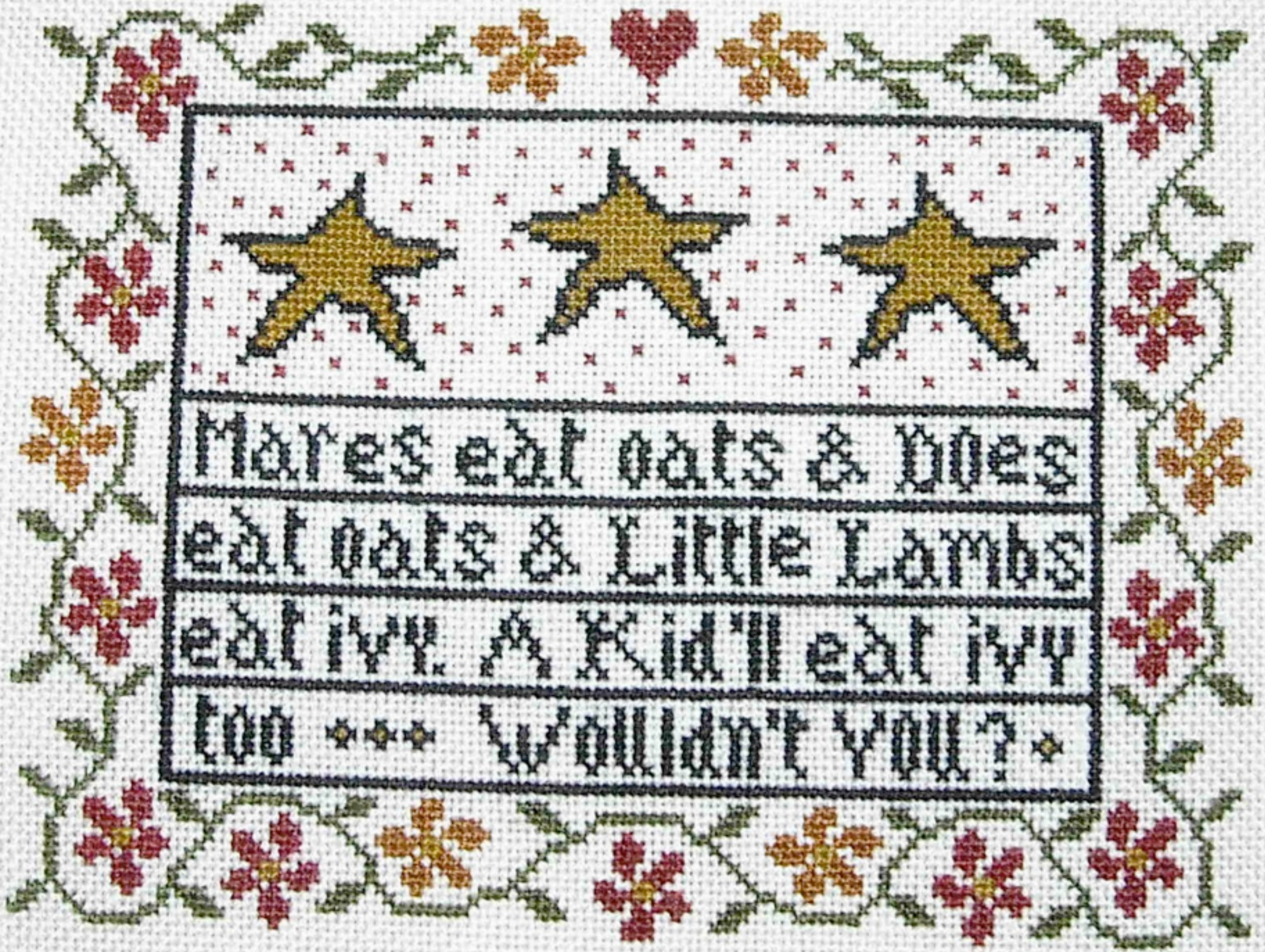
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Stanley
Spadowski's

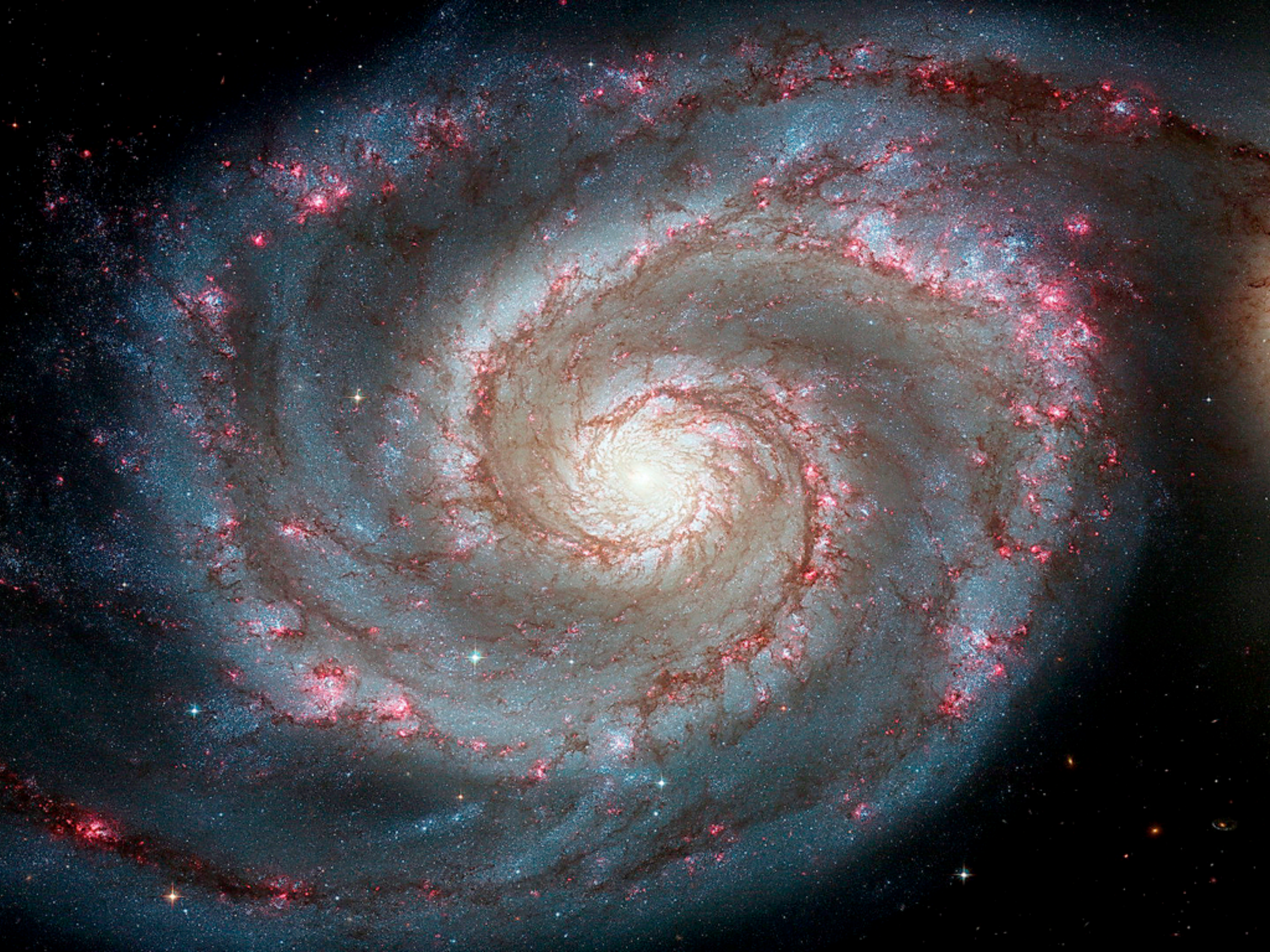

Spadowski's


WHARRGARBL

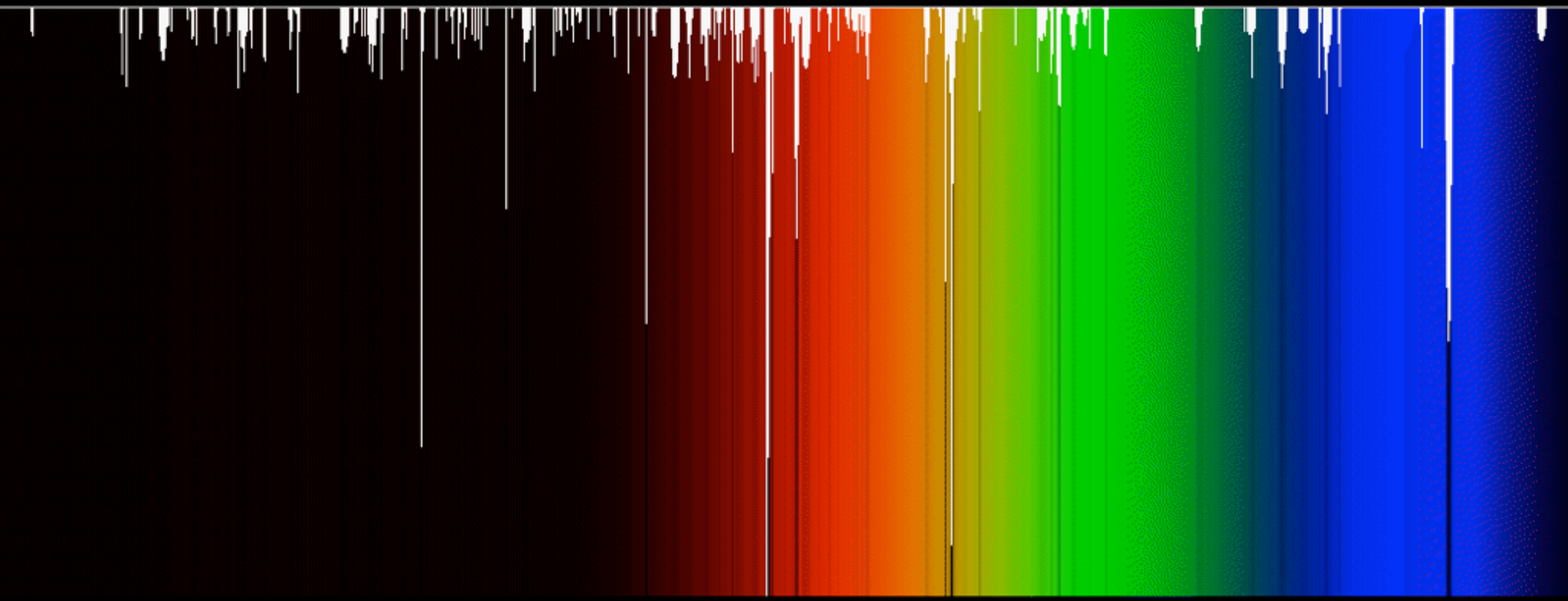

mares eat oats & does
eat oats & little lambs
eat ivy, a kid will eat ivy
too wouldn't you?







The Diffuse Interstellar Bands



. Jenniskens, F.-X. Desert

de Broglie's c

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\Rightarrow \frac{1}{\lambda} = \frac{f}{c} = \frac{m_o v}{h} \cdot \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}, \quad \because v \to c$$

$$\Rightarrow \therefore f = \frac{m_o}{\sqrt{1 - \frac{v^2}{c^2}}} \cdot h \quad \because vc = \frac{m_o}{\sqrt{1 - \frac{v^2}{c^2}}} c^2, \; \because$$

$\boxed{E = hf}$, a classic quantum energy relationship

$$\Rightarrow p = \frac{h}{\lambda} = \frac{hf}{c} = \frac{E}{c}$$, from de Broglie above and $\frac{1}{\lambda} = \frac{f}{c}$

$\Rightarrow \therefore \boxed{p = \frac{E}{c}}$, another classic quantum relationship which by special relativity is also derived as follows in two ways:

(1) $m = \frac{E}{c^2} \Longrightarrow p = mv = \frac{E}{c^2} v = \frac{E}{c}$, as $v \to c$

(2) $E^2 - c^2 p^2 = (m_o c^2)^2 \Longrightarrow E^2 = c^2 p^2 + (0)^2$, since the photon's rest or proper mass m_o

as $v \to c$

$\Rightarrow p = \frac{E}{c}$

see: Some Quantum Consequences of $E = mc^2$

Therefore,

$E = hf = h \cdot \frac{c}{\lambda} \Longrightarrow \lambda = \frac{hc}{E} = \frac{h}{mv} = \frac{h}{mc}$, as $v \to c \Longrightarrow \therefore E = mc^2$

Have Fun





